



Non-consumptive effects of odonate predators on pollinators of water lilies

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Abstract

Plant-pollinator interactions are shaped not only by environmental conditions but also by biotic interactions such as predation. Predators influence prey through direct consumption and non-consumptive effects (NCEs) arising from perceived predation risk altering prey traits such as foraging behaviour. Emergent freshwater macrophytes such as the yellow water lily (*Nuphar lutea*) create a shared interface where predators and pollinators co-occur. Yet, NCEs of predation risk on pollinators and their consequences for plant-pollinator interactions remain severely understudied, especially at aquatic-terrestrial transition zones. Therefore, we experimentally tested how artificial dragonflies, as standardised visual predation cues, shape the foraging behaviour of pollinators on *N. lutea* and assessed whether these behaviours were related to natural variation in odonate abundance. We quantified visitation rates and visit durations under artificial dragonfly, neutral object, and control treatments, and related visit duration to natural odonate abundance. Our results showed that artificial dragonfly-like cues reduced visitation rates by up to 50% and visit durations by up to 18 s per visit, across pollinator taxa. Moreover, visit duration showed a negative association with natural odonate abundance. Structural equation models revealed that pollinator behaviour was indirectly associated with environmental conditions (i.e., wind and overcast weather) via their negative effects on odonate abundance. Consequently, our findings demonstrate that predator-mediated NCEs can play a substantial role in shaping pollinator foraging on *N. lutea*, highlighting that plant-pollinator interactions at freshwater-land ecotones may depend on dynamic predation risks and underlying environmental drivers. These insights are particularly important for understanding macrophyte-insect interactions because communities at aquatic-terrestrial interfaces are especially vulnerable to shifts in environmental conditions on land and water.

Keywords Dragonfly · Aquatic-terrestrial linkage · Landscape of fear · Yellow pond lily · Trait-mediated Effects · Macrophyte

Introduction

Plant-pollinator interactions are multifaceted, as plants serve not only as sites for pollination and shelter, but pollinators themselves can function both as mutualists and as prey (Benoit and Kalisz 2020). Predation on these pollinators

influences their behaviour, survival, and consequently also their ecological roles (Romero et al. 2011; Benoit and Kalisz 2020). Predator-prey interactions have often been evaluated for direct, consumptive effects between predator and prey species. However, over the last decades, the role of “non-consumptive effects” (NCE) have gained recognition in experimental studies (e.g., Romero et al. 2011). Such effects arise from perceived predation risk, rather than direct consumption events. NCEs include alterations in prey phenotypes, such as changes in behaviour, morphology, physiology, or life history (Werner and Peacor 2003; Sheriff and Thaler 2014). By imposing fitness costs associated with defensive strategies, NCEs can equal or even exceed the population-level impacts of direct consumption (Preisser et al. 2005). This highlights the importance of understanding

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how prey perceive and respond to spatial heterogeneity in predation risk.

Non-consumptive effects of predators are often mediated through the “landscape of fear”, a conceptual framework describing how spatial variation in perceived predation risk shapes prey behaviour (Bleicher 2017; Gaynor et al. 2019). Aspects in the physical landscape and spatial differences in predation risk create zones of higher and lower perceived danger (Gaynor et al. 2019). In a landscape of fear, prey adjust their behaviour through changes in vigilance or energy allocation due to the perception of direct and indirect cues, learned behaviour, and the balance between the cost and benefit of responses. For foraging prey species such as many pollinators, a landscape of fear may cause shorter floral visits and avoidance of flowers posing high perceived risk of predation, thus decreasing foraging efficiency (Preisser et al. 2005; Laundré et al. 2010). In fact, a recent study showed that flower-visiting insects often face predation risk while foraging on flowers, driving behavioral adaptations that enable them to recognize and avoid predators or other unfamiliar objects on flowers (Antiqueira et al. 2025). These behavioural changes can have cascading effects on other species via trait-mediated-indirect-interactions (TMIIs) (Knight et al. 2005). In TMIIs, a predator exerts indirect influence on a third species by altering traits, such as pollination behaviour, of an intermediate species which then can indirectly have negative consequences for interacting species such as flowering plants (Gonçalves-Souza et al. 2008; Wirsing et al. 2021). For instance, in three-way interactions within pollination networks, predators may disrupt the behaviour of pollinating insects (intermediate species) through perceived predation risk, thereby reducing the pollination success of plants that depend on plant-pollinator interactions (Knight et al. 2005). Although non-consumptive effects are increasingly recognised as important drivers of ecological interactions, their influence on pollination dynamics remain poorly understood (Benoit and Kalisz 2020; Camurça et al. 2024).

Yellow water lilies (*Nuphar lutea*) exhibit a particularly interesting ecology and pollination strategy (Ervik et al. 1995). As a freshwater macrophyte, most of its biomass is submerged; however, through its floating leaves and emergent reproductive organs, it inhabits an aquatic-terrestrial transitional zone that links flying insect pollinators with freshwater habitats (Ervik et al. 1995). Underwater, *N. lutea* provides habitat for numerous aquatic insect larvae, including predatory odonate larvae, which use macrophyte shoots for shelter or hunting grounds (Hunger 1998; Brito et al. 2021; Sentis et al. 2022). At the water surface, its floating leaves and emergent flowers further offer suitable habitats for larval emergence (Hunger 1998) and oviposition (Röbelen and Schütte 2020). During spring and summer, also pollinators benefit from the nectar and pollen of the

brightly coloured blossoms. *N. lutea* is mainly visited by inconspicuous pollinators of the groups Diptera (including syrphid-flies) and Hymenoptera (e.g., *Apis mellifera* and *Bombus terrestris*). Fly species of the families Ephydriidae (e.g., *Notiphila*, *Hydrellia*), Scatophagidae (especially the yellow-water-lily-fly *Hydromyza livens*), and related taxa exhibit close associations with *Nuphar* species and contribute substantially to their pollination (Van der Velde and Brock 1980, Lippok and Renner 1997). *N. lutea* displays protogynous flower development, as the sexual phases of the floral organs are temporally separated, which represents an evolutionary adaptation to insect pollination and effectively reduces the likelihood of self-pollination (Lippok and Renner 1997). Although widespread, *N. lutea* remains an understudied freshwater macrophyte, which offers an ideal system to investigate non-consumptive interactions between predators and pollinators at the interface between land and water.

Odonates (dragonflies and damselflies) are some of the most effective insect predators, with exceptionally high predation success rates of up to 97% (Olberg et al. 2000). As visually oriented sit-and-wait and patrol predators, they can influence pollination not only through direct consumption of pollinators (Benoit and Kalisz 2020; Sentis et al. 2022) but potentially also by generating spatially structured predation risk that alters pollinator behaviour. Such risk effects place odonates unequivocally within a landscape-of-fear framework, where perceived predation risk can mediate trait-based responses in pollinators, potentially leading to indirect effects on plant-pollinator interactions (Bleicher 2017; Gavini and Quintero 2024). Through their complex life cycle and dual role as aquatic larvae and terrestrial adults, odonates also link freshwater and terrestrial ecosystems, making them particularly relevant for studying predator-mediated interactions at ecosystem boundaries (Knight et al. 2005). Their presence around emergent aquatic macrophytes, such as *N. lutea*, creates a natural context in which predation risk experienced by pollinators may cascade across habitats. Despite this potential, non-consumptive effects of odonate predation risk on pollinator behaviour, and the resulting consequences for plant-pollinator interactions, particularly at aquatic-terrestrial interfaces, remain poorly understood (Benoit and Kalisz 2020). This knowledge gap is especially relevant given that interfaces between freshwater and terrestrial systems are highly sensitive to environmental change, as they are affected by shifts in both systems (McFadden et al. 2023; Vagheei and Boano 2025).

Here, we examined how visually perceived predation risk from odonates shapes pollinator behaviour in an aquatic plant-pollinator system. Using artificial dragonfly models as standardised visual predation cues, we tested their effects on pollinator visitation rate and visit duration across

different pollinator taxa visiting *N. lutea*. To place these experimental effects within the broader ecological context of natural predation risk, we additionally quantified natural odonate abundance in the field. We then used a structural equation modelling (SEM) approach to evaluate how abiotic conditions, including wind and weather, influenced natural odonate abundance and how this variation indirectly relates to pollinator visit duration. Specifically, we expected that (i) artificial dragonflies would reduce both pollinator visitation rate and visit duration relative to control conditions; and that (ii) higher natural odonate abundance would be associated with shorter visit durations. By integrating experimental manipulation with observational data, our study examines non-consumptive predator effects within a landscape-of-fear framework and assesses their potential to generate trait-mediated indirect interactions (TMIIs) affecting plant-pollinator interactions. Together, these results provide insight into non-trophic predator-prey interactions, highlight the role of abiotic drivers in shaping perceived predation risk, and improve our understanding of cross-ecosystem linkages and pollination dynamics at aquatic-terrestrial boundaries, with potential consequences for the reproductive ecology of *N. lutea*.

Materials and methods

Study site

We conducted the field experiment in an urban pond in Hamburg, Germany (53°33'33.9"N 9°51'41.9"E). This artificially created pond (maximum depth=0.6 m) is surrounded by abundant riparian vegetation and harbours various freshwater macrophyte species (e.g., *Myriophyllum spicatum*, *Ceratophyllum demersum*, *Nuphar lutea*, *Phragmites australis*). There was a high abundance of floating-leaved plants, mainly *N. lutea* with a surface coverage of 90% during peak flowering season in June/July 2025. At the interface between land and water, floating-leaved flowering plants such as *N. lutea* function as ecological connectors between aquatic and terrestrial systems. The pond harboured different species of fish, most notably of the genus *Cyprinus* (carp), as well as water birds such as *Gallinula chloropus* (common moorhen) and *Anas platyrhynchos* (mallard). We characterised the water physicochemical conditions (HACH HQ Multiprobe) in June, July and August: the water temperature ranged from 19 to 22 °C during the day, pH ranged from 7.2 in August to 7.7 in June and July, conductivity ranged between 408 and 428 µS/cm, and dissolved oxygen ranged from 6.4 to 8.2 mg/L which corresponded to 73–83% dissolved oxygen.

Study design

To investigate possible effects of artificial dragonflies on pollinator behaviour (visitation rate, visit duration), we performed a field experiment using a randomised block design (Fig. 1a). We observed a total of 22 one-hour blocks over a span of 15 different days between June and August. This corresponds to an observational period of 1320 min. We selected nine yellow water lily flowers per block with an average spacing of 1 m (max: 2 m, min: 0.5 m) between flowers and observed visits by pollinating insects. Visit duration was measured using a stopwatch. We assigned the treatments randomly to individual flowers, resulting in three flowers per treatment (artificial dragonfly, neutral object, control). To examine a potential relationship between natural odonate abundance and visit duration, we recorded the number of odonates active in the immediate vicinity of a flower during each pollinator visit (individuals/m³). In total, we recorded 176 odonates over 459 plant-pollinator interactions with a sum of 449 min of visitation. We considered that *N. lutea* follows a protogynous flower development and thus, initially undergo a female phase, followed by a male phase lasting several days. We selected the flowers at the same stage of development and at equal distances from each other. Since individual yellow water lily flowers only bloom for a few days before sinking and being replaced by new flowers, it was not possible to ensure a perfectly even distribution of flowers over a predetermined area throughout the entire flowering period. However, we minimised environmental differences between blocks as much as possible to ensure the most independent distribution and observation of the treatments. Wind and weather were recorded as binary environmental variables with “wind” as ~2 m/s or higher or “no wind” when calm. We recorded weather as either overcast or clear as these were the only conditions where pollinators of *N. lutea* were effectively active.

Visual predation cues

To test the effects of perceived dragonfly predation risk of pollinators of *N. lutea*, we attached artificial dragonflies with a length of 5.5 cm and a maximum width (wingspan) of 6 cm and made from PVC plastic on the fully emerged inflorescences of *N. lutea* (Fig. 1b). The artificial dragonflies display the most important morphological characteristics of Anisoptera and were comparable in dimensions to four-spotted chasers (*Libellula quadrimaculata*), a commonly found dragonfly species at the study site during the observational period. We attached the artificial dragonflies to the side of the water lily flowers as field observations have previously shown that the dragonflies found there (mainly *Libellula quadrimaculata*) use the *Nuphar* flowers

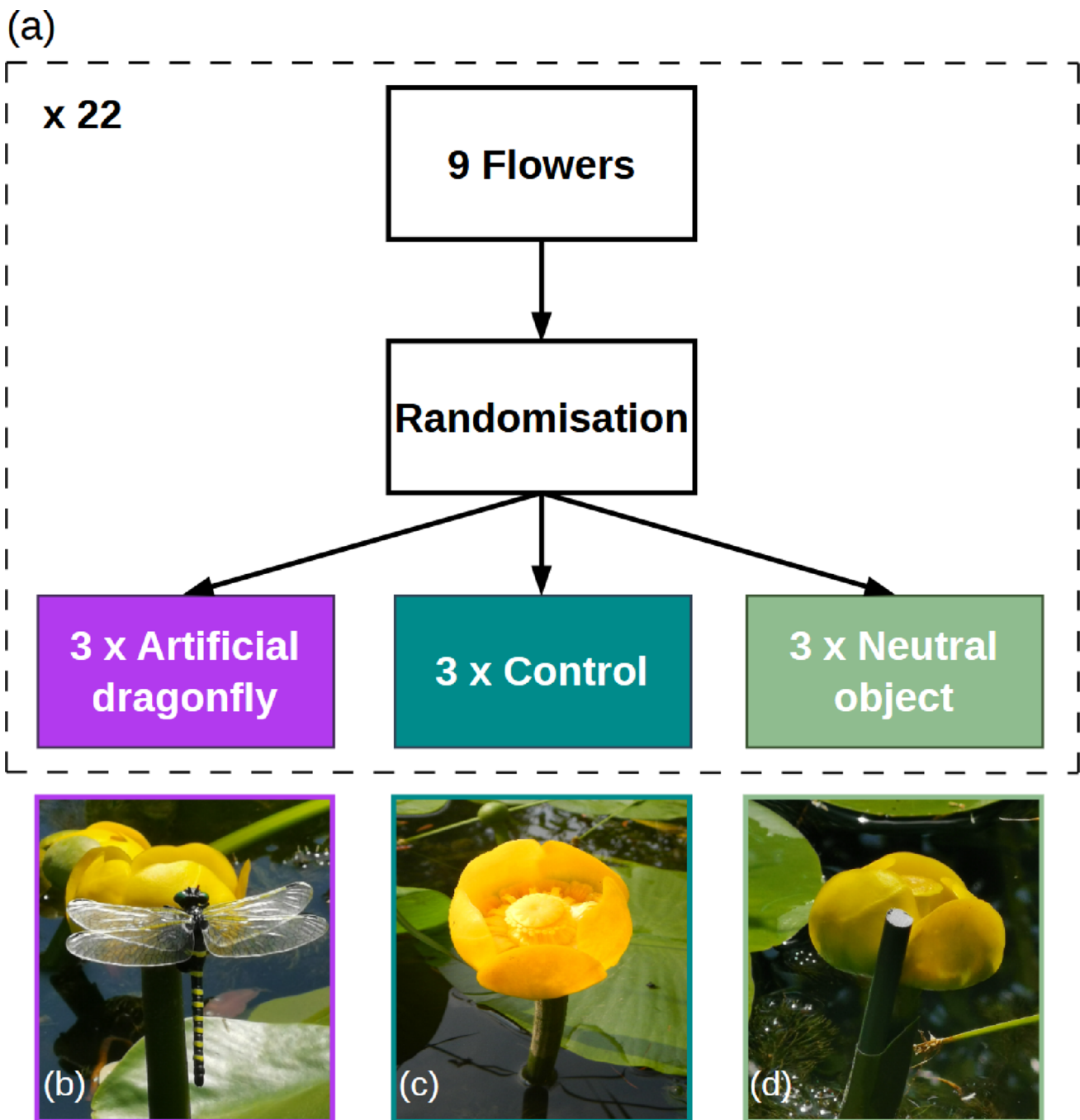


Fig. 1 Study design. **a** Conceptual diagram of block design of the experiment, **b** flower with artificial dragonfly treatment, **c** control flower without any objects attached, **d** flower with neutral object treatment

and upright leaves as perching points. The artificial dragonflies were placed in such a way as to prevent mechanical disturbance to pollinators.

To control for a possible influence on pollinating insect behaviour caused merely by the presence of a foreign object on the flowers, black PVC plastic rods were used as a counter-treatment (neutral object) (Fig. 1d). These rods measured also 6 cm in length and 0.6 cm in diameter. They were

attached to the flowers in positions comparable to those of the artificial dragonflies, using unobtrusive green adhesive tape applied to the stem. Through this neutral-object treatment, we could differentiate between the effects specifically attributable to dragonfly-like morphological characteristics and a foreign object.

Statistical analysis

We performed all statistical analyses using R Version 4.5.1 (R Core Team 2025). All data and code to fully reproduce our analyses are publicly available under: <https://doi.org/10.5061/dryad.1c59zw4bt>.

To assess how experimental treatments influenced visitation rates (visits per treatment per hour), we used generalized linear mixed models (GLMM) using the package lme4 (Bates et al. 2015). We constructed the separate models for Diptera and Hymenoptera and with observational blocks as random effect and with treatment as fixed effect. For each treatment inside each block, we aggregated count data, including zero visits. We constructed the models following a Poisson and a negative binomial distribution and tested them for overdispersion and zero inflation. Overdispersion was not present in the Diptera model and negligible in the Hymenoptera model. We compared models with Poisson and with negative binomial distribution using AIC and Likelihood Ratio Test, and because the models were similar ($\Delta \text{AIC} < 2$; $\text{LRT: } X^2 = 0.0011$, $p = 0.97$), we used Poisson GLMMs. To assess the normality of error distributions and homoscedasticity of the models, we visually inspected diagnostic plots such as residuals vs. fitted values. We assessed contrasts among treatments using estimated marginal means (EMMs) with the package emmeans and a Tukey correction for multiple comparisons (Lenth et al. 2025). Blocks in which no Hymenoptera were observed throughout the sampling period were excluded from the Hymenoptera analyses because they represented periods without any activity of the focal pollinator group, and therefore contained no information on treatment effects. Hence they would have inflated standard errors and lowered estimated marginal means, without explaining effects due to our experimental manipulations (Diptera: 22 blocks, Hymenoptera: 16 blocks). To compare treatment effects between pollinator groups, we constructed models including both groups, comparing models with and without interaction between treatment effects and pollinator group using Likelihood Ratio Tests.

To assess how different experimental treatments influenced visit durations, we also used linear mixed models using the package lme4 (Bates et al. 2015). Again, we constructed the models with observational blocks as random effect and with treatment and odonate abundance as fixed effects. We compared models with and without interaction between treatment and odonate abundance using Likelihood Ratio Tests. We obtained minimal adequate models by excluding factors, as well as interactions without significance and by using AIC comparisons through a top-down approach, starting with full models. We applied a log10-transformation to the response variable (visit duration) prior to testing for normal distribution. We assessed

the normality of errors and homoscedasticity of the models visually using residuals vs. fitted values plots. We assessed contrasts among treatments using estimated marginal means (EMMs) with the package emmeans and a Tukey correction for multiple comparisons (Lenth et al. 2025). To compare treatment effects between pollinator groups, we constructed a third model using data from both groups using the same procedure, comparing models with and without interaction using Likelihood Ratio Tests.

To examine a potential relationship between odonate abundance and pollinator visit duration, we calculated Pearson coefficients for each pollinator group using log10-transformed data for visit duration. We then visualised possible interactions between treatment and odonate abundance regarding visit duration using predicted values from the LMMs.

To test indirect associations between visit duration and environmental factors, we constructed structural equation models (SEMs). We constructed the path models using the package lavaan (Rosseel 2012). We included indirect effects of both wind and weather mediated by odonate abundance. Exogenous variables are allowed to correlate by default, so wind and weather were modelled with a freely estimated covariance. Consequently, each path coefficient reflects the effect of a predictor while statistically controlling for the other (Rosseel 2012). We transformed the data of the response variable (visit duration) using log10 to achieve a normal distribution. We used Most Likelihood as an estimator for the models. We constructed the path models separately for both dipteran and hymenopteran pollinators. We evaluated model fit using multiple fit indices (X^2 , RMSEA, SRMR, CFI, TLI). Following recent recommendations, we interpreted fit indices descriptively rather than using strict cut-off values and considered theoretical expectations and path coefficients (Groskurth et al. 2024). Detailed hypotheses for each included link are presented in Table S1.

Results

Pollinators of *Nuphar lutea*

We found a diverse assemblage of insect taxa, with Diptera ($n = 281$) representing the most frequent pollinator-group on *N. lutea*. Dipterans included generalist taxa such as hoverflies (Syrphidae), including *Episyrphus balteatus*, as well as taxa associated with specialists of *Nuphar*, notably Ephydriidae (specifically Hydrelliinae) and Scathophagidae, with individuals tentatively identified as *Hydromyza livens*. Hymenopterans ($n = 162$) included the honeybee (*Apis mellifera*) and the buff-tailed bumblebee (*Bombus terrestris*). We observed Coleopteran visitors only infrequently ($n = 15$)

and therefore excluded them from further analyses. In addition to pollinators, we recorded several odonate taxa at the study site, including damselflies of the family Coenagrionidae, with the small red-eyed damselfly (*Erythromma viridulum*) being the most abundant odonate species overall. We also observed dragonflies of the genera *Aeshna*, *Libellula*, and *Sympetrum*, including *Libellula quadrimaculata* and *Sympetrum sanguineum*, with *Libellula quadrimaculata* being the most commonly observed dragonfly species.

Treatment effects on pollinator visitation rates

Our artificial dragonflies significantly lowered the mean visitation rates (EMMs) of both dipterans and hymenopterans compared to the control in our study system. The mean visitation rate of dipterans (Fig. 2a) was estimated to be 50% lower on flowers with artificial dragonflies (rate: 2.29, CI 1.51–3.47) than on control flowers (rate: 4.54, CI 3.10–6.65; $p < 0.01$). The mean visitation rate of hymenopterans

(Fig. 2b) was estimated to be 46% lower on flowers with artificial dragonflies (rate: 1.47, CI 0.796–2.71) than on control flowers (rate: 2.74, CI 1.55–4.87; $p < 0.01$). Difference in EMMs of dipteran visitation rate was significant between neutral objects and control flowers, but non-significant in hymenopterans. The mean visitation rate of dipterans was estimated to be 33% lower on flowers with neutral objects (rate: 3.06, CI 2.05–4.56) than without ($p = 0.01$). Differences in mean visitation rates (EMMs) between artificial dragonflies and neutral objects was non-significant in either pollinator group. Differences regarding treatment effects were non-significant between pollinator groups (LRT: $X^2 = 0.87$, $p = 0.65$).

Treatment effects on pollinator visit duration

Dragonfly-like visual cues significantly shortened the mean visit duration of both dipterans and hymenopterans compared to the control. Floral visits of dipterans (Fig. 3a) were

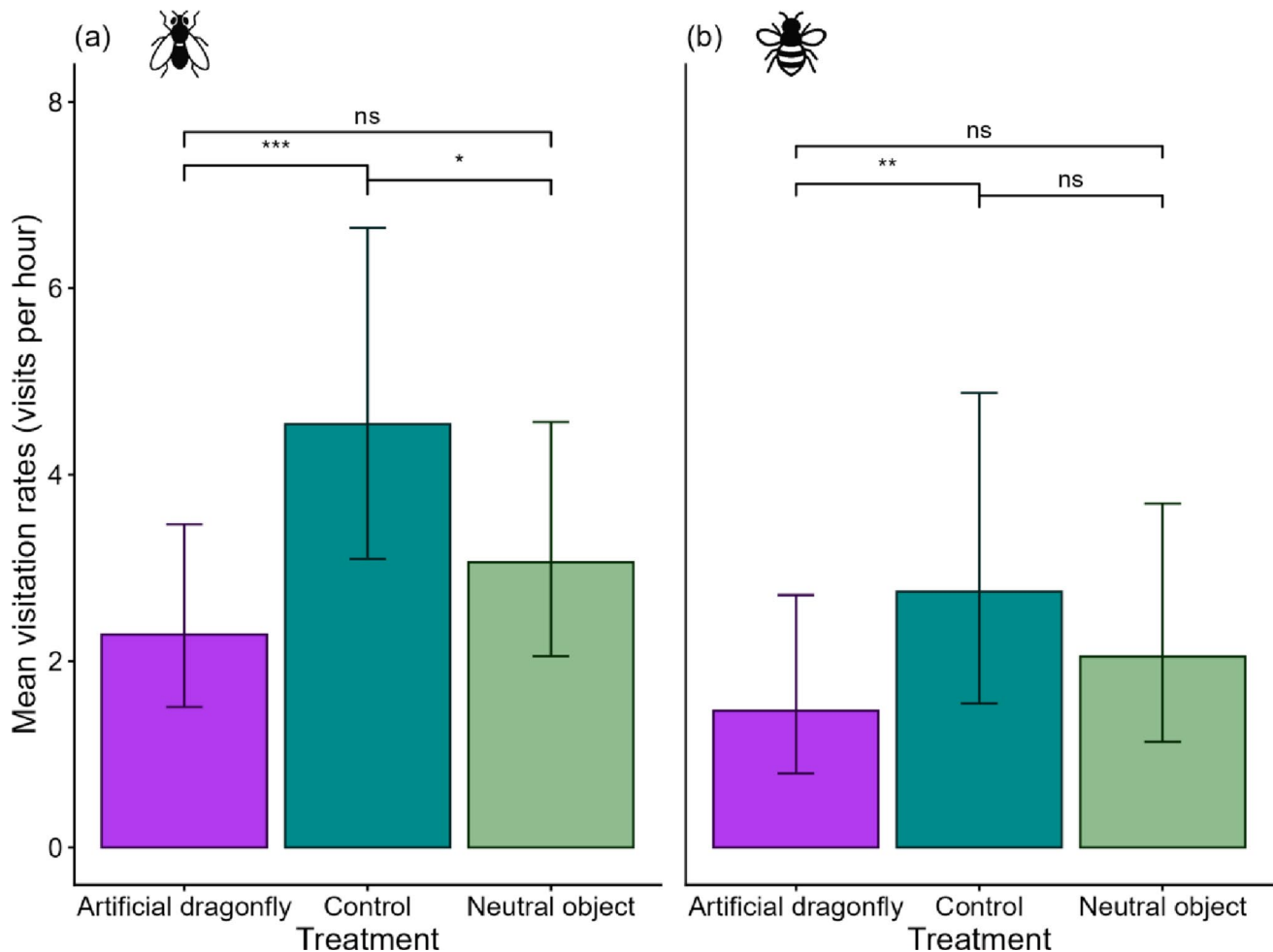


Fig. 2 Mean visitation rates (back-transformed EMMs, visits per hour) of flowers with either artificial dragonflies, neutral objects, or no manipulation (control), for **a** Diptera ($n=281$) and **b** Hymenoptera ($n=162$). Statistical differences between treatments were marked

with asterisks: ($\alpha=0.05$), *** = very high significance ($p < 0.001$), ** = highly significant ($p < 0.01$), * = significant ($p < 0.05$), non-significant (ns)

estimated to be 38% (Δ 18.35 s) shorter on flowers with artificial dragonflies (emm: 29.51 s, CI 22.90–37.15 s) than on control flowers (emm: 47.86 s, CI 39.81–58.88 s; $p < 0.01$). Floral visits of hymenopterans (Fig. 3b) were estimated to be 40% (Δ 15.80 s) shorter on flowers with artificial dragonflies (emm: 23.99 s, CI 16.98–33.11 s) than on control flowers (emm: 39.80 s, CI 30.20–52.48 s; $p < 0.01$). Difference in EMMs of dipteran visit duration was significant between neutral objects and control flowers, but non-significant in hymenopterans. Floral visits of dipterans were estimated to be 24% (Δ 11.55 s) shorter on flowers with neutral objects (emm: 36.31 s, CI 29.51–44.67 s) than on those without (emm: 47.86 s, CI 39.81–58.88 s; $p = 0.04$). Difference in EMMs of visit duration between artificial dragonflies and neutral objects was non-significant in either pollinator group. Treatment effects regarding visit duration were not significantly different between dipterans and hymenopterans (LRT: $X^2 = 1.85$, $p = 0.40$).

Relationship between visit duration and natural odonate abundance

Visit duration showed a negative association with natural odonate abundance in both dipteran and hymenopteran pollinators (Fig. 4). The LMMs revealed a significant interaction between treatment and natural odonate abundance effects on visit duration, that was present only for dipterans (LRT: $X^2 = 7.85$, $p = 0.02$) but not for hymenopterans (LRT: $X^2 = 0.12$, $p = 0.94$).

Structural equation models (SEMs)

The SEMs confirmed the previous results in that the artificial dragonflies significantly decreased visit duration in both pollinator groups and neutral objects lowered visit duration significantly in dipterans (Fig. 5a) but not in hymenopterans (Fig. 5b). Moreover, visit duration showed a negative correlation with natural odonate abundance of similar magnitude

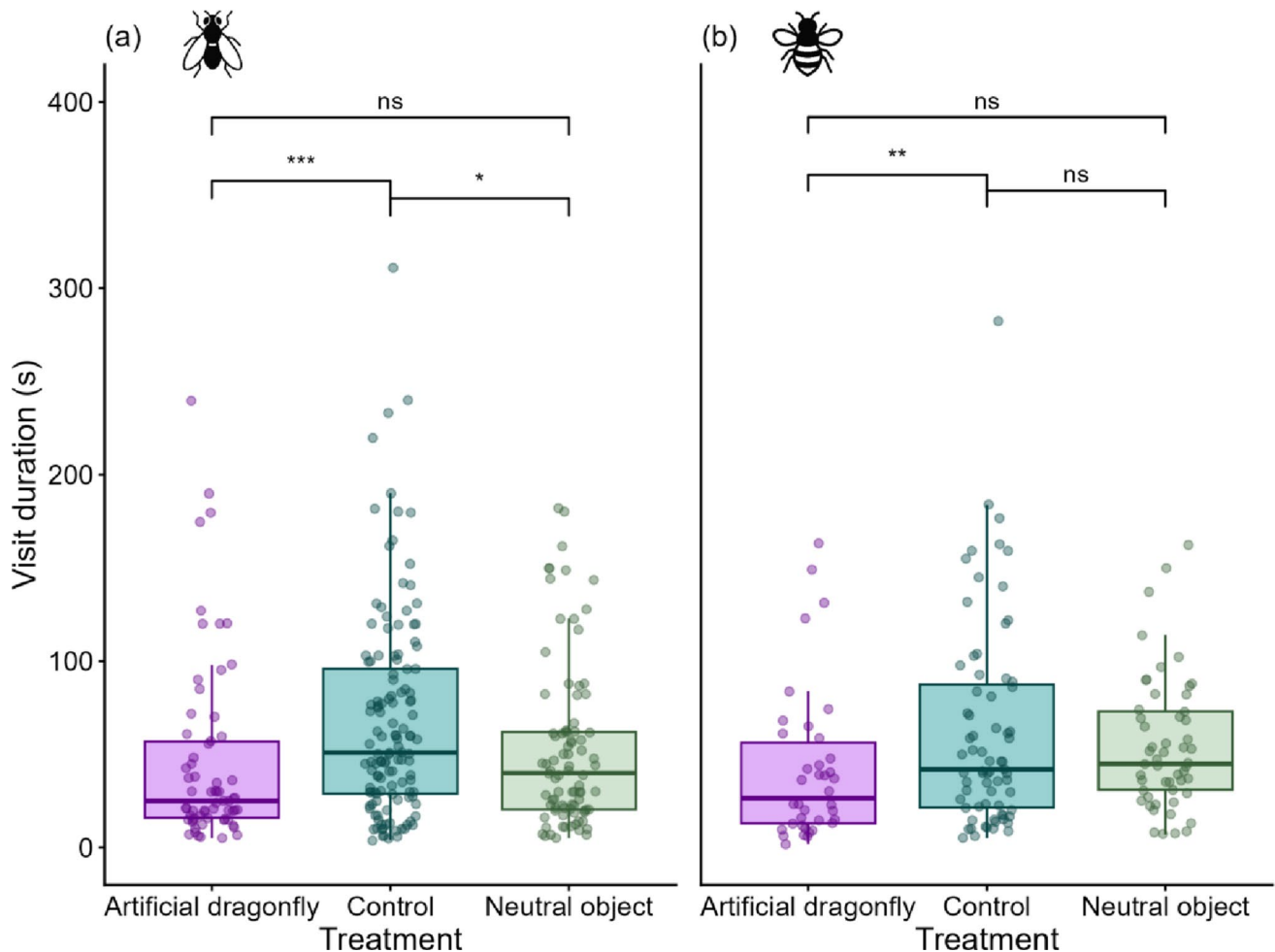


Fig. 3 Visit duration (time in seconds) of **a** Diptera ($n = 281$) and **b** Hymenoptera ($n = 162$) on flowers with either artificial dragonflies, neutral objects, or no manipulation (control). Each point represents the duration an observed insect has visited a *N. lutea* flower. Statistical

differences between treatments were marked with asterisks: ($\alpha = 0.05$), *** = very high significance ($p < 0.001$), ** = highly significant ($p < 0.01$), * = significant ($p < 0.05$), non-significant (ns)

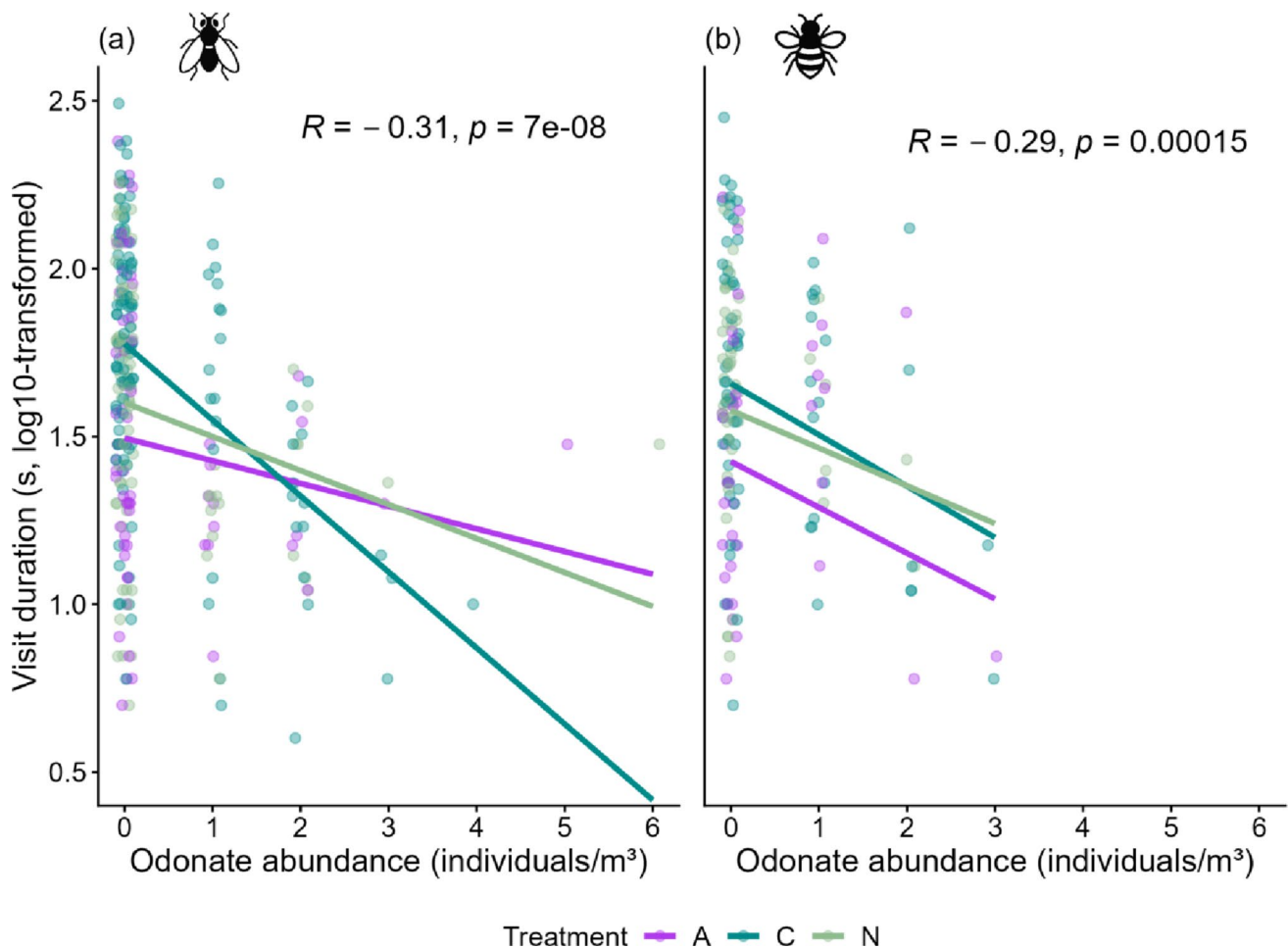


Fig. 4 Relationship between visit duration (s, log10-transformed) and natural odonate abundance (individuals/m³) for **a** Diptera ($n=281$) and **b** Hymenoptera ($n=162$). To aid interpretation, a small amount of noise has been added to each point on the x-axis. Coloured lines represent predicted values for treatments (artificial dragonfly (A), con-

trol (C), neutral object (N)) from linear mixed effects models (LMMs) representing the interaction between treatment and natural odonate abundance. Significance levels of interaction: Diptera (*), Hymenoptera (ns). See Figure S1 for untransformed values

in both pollinator groups. However, the SEMs additionally show direct and indirect relationships with environmental factors. Wind and clear weather exhibited a moderate negative correlation in both path models (Diptera: $r = -0.31$; Hymenoptera: $r = -0.42$). Natural odonate abundance was negatively associated with wind and positively associated with clear weather in both models. However, the relationship between environmental factors and odonate abundance was weaker during observations of hymenopterans. Path models revealed indirect relationships between visit duration and environmental factors, mediated by odonate abundance. For dipteran pollinators (Fig. 5a), visit duration showed a positive indirect association with wind and a negative indirect association with clear weather. For hymenopteran pollinators (Fig. 5b), indirect associations were non-significant. Overall, visit duration showed slightly stronger responses in Diptera (standardised: -0.68 ; $R^2 = 0.148$) than in Hymenoptera (standardised: -0.53 ; $R^2 = 0.120$). Detailed model fit

indices for both structural equation models can be found in Table S2. Overall, fit indices indicated stronger correspondence between the model and the data for Diptera (Fig. 5a) than for Hymenoptera (Fig. 5b).

Discussion

Pollinator behaviour can be altered by predation risk through NCEs (Dukas 2001; Dukas and Morse 2003; Knight et al. 2005), disrupting plant-pollinator mutualisms with cascading effects on plant reproduction (Werner and Peacor 2003; Knight et al. 2005; Gavini and Quintero 2024). Our results show that dragonfly-like visual cues of predation risk, simulated by artificial dragonflies, elicited behavioural responses consistent with non-consumptive predator effects on pollinator behaviour on *Nuphar lutea*. To the best of our knowledge, this is the first study observing the influence of

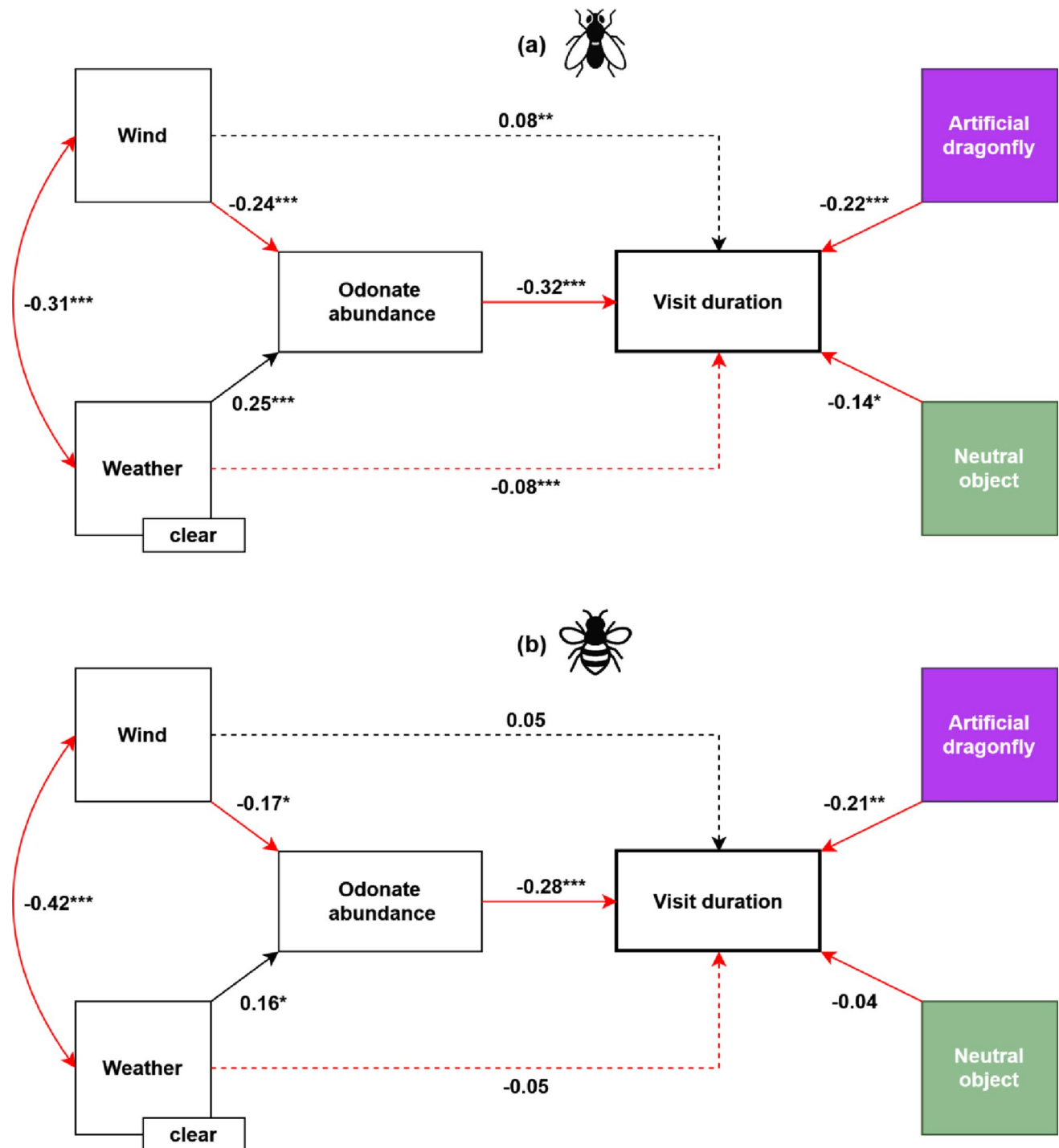


Fig. 5 Structural Equation Model for **a** Diptera ($n=281$) and **b** Hymenoptera ($n=162$). Direct effects are represented by solid paths, while indirect effects are represented by dashed paths. Paths from weather refer to “clear” weather with overcast weather as a reference. Paths from wind refer to wind (≥ 2 m/s) with no wind as a reference. Paths from artificial dragonfly and neutral treatment are each compared to

control (reference). Red paths indicate negative effects, while black paths indicate positive effects. Values are standardised path coefficients (scale: -1 to $+1$). Statistical significance of effects was marked with asterisks: ($\alpha=0.05$), *** = very high significance ($p<0.001$), ** = highly significant ($p<0.01$), * = significant ($p<0.05$)

artificial dragonflies on pollinators of macrophytes. Both visitation rate and visit duration were significantly reduced on flowers treated with artificial dragonflies compared to control flowers in our study pond. The effects on pollinator behaviour were consistent across taxa and occurred independently in Hymenoptera and Diptera. Moreover, structural equation models revealed that pollinator behaviour was indirectly associated with environmental conditions through their association with odonate abundance. Together, these novel findings support the hypothesis that odonates can exert NCEs on pollinators at aquatic-terrestrial interfaces and imply the need to consider how environmental conditions indirectly influence these insect-macrophyte dynamics.

The strong reduction in visitation rates under the artificial dragonfly treatment for both dipterans and hymenopterans indicate that dragonfly-like visual predator cues alone can significantly deter floral visitors. Our results show that artificial dragonflies reduced the visitation rate of pollinators by 50% (Diptera: Fig. 2a) and 46% (Hymenoptera: Fig. 2b). These findings exceed the average 36% reduction in pollinator visits under elevated predation risk, as reported in a meta-analysis, and may potentially have severe consequences for plant fitness through TMIs (Romero et al. 2011). While only restricted to one studied pond, our findings align with previous studies showing that pollinators are highly responsive to visual signals associated with predation, such as crab spiders (Gonçalves-Souza et al. 2008). Odonates employ a wide variety of hunting strategies that differ among species. However, this behavioural trait diversity of odonates can also depend on environmental contexts, such as land use (De Resende et al. 2021). While some species hunt as sit-and-wait predators from concealed perches within riparian or emergent vegetation, others are highly active aerial predators that capture prey almost continuously while in flight (Sentis et al. 2022; De Resende et al. 2021). During the experiment, we visually observed multiple dragonfly species within the family Libellulidae, that used the flowers of *N. lutea* as perching points. Libellulidae are known to include numerous perching species (Olberg et al. 2000); a behavioural response by pollinators to the artificial dragonflies may therefore plausibly reflect a common spatial avoidance pattern due to sit-and-wait predators. Interestingly, the neutral-object treatment also caused a moderate reduction in dipteran visits, though not in hymenopterans. This implies that dipterans may be more sensitive to structural disturbances or novel objects, whereas hymenopterans appear to differentiate between biologically relevant and irrelevant cues more effectively. Although such taxon-specific behavioural plasticity in response to visual stimuli has been reported across pollinator groups (Romero et al. 2011), we found no evidence for overall differences in responses

between pollinators of *N. lutea*, as artificial dragonflies reduced visitation rates similarly across pollinator groups. A more taxonomically precise quantification of pollinator groups or individual species would likely have revealed finer differences in behavioural patterns both within and between Diptera and Hymenoptera. The non-significant difference between artificial dragonfly and neutral-object treatments in visitation rates suggests a partial contribution of unspecific mechanical or visual disturbance (neophobia: Antiqueira et al. 2025). However, the consistent and stronger reduction under artificial dragonflies suggests that most of the response is attributable to risk perception rather than an artefact of the study design.

Reduced visit durations on flowers with artificial dragonflies reinforce the idea that perceived predation risk affects not only the decision to approach but also the time spent foraging. Our results showed that artificial dragonflies decreased the mean visit duration of pollinators by 38% (Diptera: Fig. 3a) and 40% (Hymenoptera: Fig. 3b), comparable to the 51% average reduction in visit duration, as reported by a meta-analysis (Romero et al. 2011). Given *N. lutea* protogynous floral phases (Ervik et al. 1995), shorter visit durations could have potentially reduced pollen transfer efficiency and thus reproductive success (Knight et al. 2005). Although we did not measure fitness directly, this possibility is consistent with findings of a meta-analysis showing that interference in plant-pollinator mutualisms were associated with an average 17% reduction in plant fitness (Romero and Koricheva 2011). To better understand the cascading effects of odonate predation risk on the reproductive success of *N. lutea*, we would need to quantify the extent to which pollen limitation may occur. Similar to visitation rates, the non-significant difference between artificial dragonfly and neutral-object treatments suggests a partial contribution of object-bound disturbance, though the consistent and stronger reduction under artificial dragonflies highlights that most of the response is attributable to risk perception. These findings parallel experimental work showing that pollinators trade-off feeding time for safety under elevated predation risk (Romero et al. 2011).

The shared aquatic-terrestrial interface created by *N. lutea* supports diverse odonate assemblages, which impose substantial predation risk on flower-visiting insects (Knight et al. 2005). Given this consistently high predation pressure around *N. lutea*, pollinators may adjust their behaviour in response to odonate abundance, relying on visual cues and movement in their surroundings. Consistent with this, visit duration showed a strongly negative relationship with odonate abundance across both pollinator groups (Fig. 4). Interestingly, the linear mixed models revealed a significant interaction between treatment and odonate abundance only present in Diptera (Fig. 4a). The significant interaction

between treatment and odonate abundance (Fig. 4a) indicates that dipterans respond to static, dragonfly-like visual cues but may be even more sensitive to other relevant stimuli like movement or olfactory cues (Romero et al. 2011). These findings align with studies demonstrating that *Drosophilid* flies exhibit diverse sensitivities to visual predation cues (MacLeod and Luong 2024) and escape through exceptionally fast motor escape skills (Card and Dickinson 2008). The spatial distribution of dipteran species that are specialised pollinators of *Nuphar* (Lippok and Renner 1997) is more restricted compared to other pollinator groups. For these dipteran species, the trade-off between resource acquisition and predator avoidance may be less pronounced than for hymenopterans like bees, which forage across larger spatial scales (Beekman and Ratnieks 2000; Huey and Nieh 2017). Hymenopterans likely face higher costs when reacting to predation risk, as avoiding certain high reward flowers or patches may considerably reduce foraging efficiency (Gavini and Quintero 2024). Although visit duration was also negatively correlated with natural odonate abundance in Hymenoptera, treatment effects remained clearly distinguishable over rising odonate abundance. Due to their larger body size, Honeybees (*Apis mellifera*) and buff-tailed bumblebees (*Bombus terrestris*) are likely more frequently preyed upon by large anisopterans (Olberg et al. 2005). Therefore, hymenopterans could react differently to perceived predation risk posed by dragonflies compared to damselflies. However, we could not disentangle possible differences since odonate abundance was recorded only at a low taxonomic resolution and a more detailed quantification might reveal more nuanced predator-specific effects on different pollinator groups. Although we were able to examine the relationship between odonate abundance and visit duration of pollinators, We were not able to draw similar conclusions about pollinator visitation rates. This would require a different approach, in which odonate abundance is recorded independently and at regular intervals throughout the observational blocks.

Our study not only shows direct effects of simulated predation risk on pollinators of *N. lutea* and associations with live odonate abundance, but also revealed indirect relationships with environmental variables, mediated by odonate abundance. These findings fit within the inherently dynamic “landscape of fear” framework, in which shifts in predator distribution over time alter the spatial patterns of perceived predation risk (Laundré et al. 2010). Environmental factors that positively influence odonate abundance may therefore intensify this “landscape of fear,” exacerbating trade-offs faced by pollinators (Gavini and Quintero 2024; Gaynor et al. 2019). Trade-offs can, for example, include the cost and benefits of increased vigilance, spatiotemporal redistribution, social behaviours, or foraging (Gaynor et al.

2019). Our findings imply that wind had a negative effect on odonate abundance in the study area, thus indirectly causing an elevation of pollinators foraging duration. As primarily ectothermic organisms, the ability to thermoregulate differs among odonate species and is mostly dependent on body size and hunting behaviour (Guerra et al. 2026). Most perching species are “thermal conformers” that rely mostly on air temperature or heliothermic species that become active when direct sunlight is available (De Marco and Resende 2002). Many large “fliers” are endothermic and produce endogenous heat with the ability to thermoregulate by controlling their haemolymph circulation (De Marco and Resende 2002). Therefore, direct sunlight can increase odonate activity and hunting behaviour, which is especially relevant in a warming world (Guerra et al. 2026). Our findings imply that clear weather has a positive effect on odonate abundance, increasing their activity and indirectly decreasing pollinator visit duration. We observed that small red-eyed damselflies became active almost immediately when clouds dispersed and direct sunlight was present. This observation is consistent with reports indicating that this species depends on warm conditions and exhibits an almost immediate reduction in activity when direct sunlight is obscured by cloud cover or during strong winds (Hunger 1998).

Many pollinators benefit from the same weather conditions (Miyashita et al. 2023) that also favour odonates (De Marco and Resende 2002). On clear, calm days, both odonate abundance and pollinator activity may increase. This conflict between favourable environmental conditions and the need to avoid predation risk can further heighten the trade-offs experienced by pollinators. Therefore, environmental variables can shape the non-consumptive effects between odonates and pollinators, consequently amplifying cascading effects on plant-pollinator mutualisms through TMIs. Simultaneous visits under high pollinator activity may have reduced observational accuracy. The path models (Fig. 5) explained 15% (Diptera) and 12% (Hymenoptera) of the variance in visit duration. The relatively low variance explained by the models may reflect the inherent complexity of pollinator behaviour, which is influenced by numerous factors beyond predation (Mailly et al. 2025). Additionally, species-specific differences, not captured due to low taxonomic resolution, may further contribute to unexplained variation. As species and their interactions become increasingly affected by climate change, it is essential to account for the extent to which environmental variables can affect ecological interactions (Vagheei and Boano 2025). In addition to trophic interactions, it is especially important to also consider such non-trophic interactions and their cascading effects (e.g., TMIs) (Benoit and Kalisz 2020; Romero et al. 2011).

In conclusion, our findings suggest that visually perceived dragonfly-like cues can elicit behavioural responses in *N. lutea* pollinators consistent with non-consumptive responses to predation risk. Additionally, our data suggests that odonates may not only influence the foraging behaviour of pollinators via direct effects, but also by mediating indirect effects of environmental conditions on pollinator behaviour. Therefore, future studies of NCEs and TMIIs should additionally consider the possible indirect effects of environmental factors. Our findings highlight relevant NCEs at aquatic-terrestrial interfaces, impacting plant-pollinator interactions and thus possibly influencing the reproductive success of *N. lutea* through TMIIs. Understanding how non-consumptive effects shape the pollination dynamics in freshwater-land ecotones could support conservation efforts aiming to preserve biodiversity on both land and water.

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Data availability All data and code to fully reproduce our analyses are publicly available under: <https://doi.org/10.5061/dryad.1c59zw4bt>.

Declarations

Competing interests The authors declare no competing interests.

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